

Vetsa Sai Vivek

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with hands-on exposure to embedded systems and IoT-based applications. Proficient in Java-based application development with a strong focus on scalable system design, role-based workflows, and data-driven decision enhancement. Passionate about building robust technology solutions that solve complex business challenges.

EXPERIENCE

Bosch Global Technologies Pvt Ltd

Associate Software Engineer

Pune, India
Mar 2025 – Present

- Currently, I am working on the design and development of a full stack Java-based enterprise web application **Order-Kraft**, a supply chain and order management platform aimed at enhancing end-to-end business operations.
- The system enables intelligent handling of order life cycle management from intake to delivery by integrating inventory tracking, production scheduling, supplier coordination, and shipment monitoring into a unified ecosystem.
- The platform leverages AI-driven forecasting to predict inventory demand and optimize stock levels, reducing operational inefficiencies while supporting real-time visibility and scalable decision-making among stakeholders.
- Involved in building scalable backend services using Spring Boot integrated with REST and SOAP APIs to enable seamless data flow across inventory, supplier, production and user management systems.
- Actively developing responsive frontend interfaces using Angular and JavaScript to enhance user interaction and system usability.
- Collaborating within Agile environments using Jira for workflow management and GitHub for version control and deployment support, enabling structured development and code sharing across teams.
- Leveraging Oracle Database for secure data handling and role-based access management while ensuring efficient system performance.

Bosch Global Technologies Pvt Ltd

Associate Engineer – Intern

Bangalore, India
Sept 2022 – May 2023

- I have worked in the Actuators and Drivers (EAD) domain supporting the development of mobility-focused automation technologies for automotive systems.
- Contributed to the software development of a Power Window Control System integrated with Anti-Pinch Safety functionality, designed to enhance passenger safety and operational reliability.
- The system enables automated window movement through motor-driven control logic while incorporating obstruction detection mechanisms to prevent injury or damage.
- This experience provided exposure to safety-critical automotive system design, embedded software logic, and real-world mobility automation solutions.

SKILLS

Programming Languages

Web Development

Frameworks & Technologies

Embedded Systems

Tools & Platforms

Database Management

Java, SQL, C Language

UI/UX Design, JavaScript

Spring Boot, Angular

Arduino, Raspberry Pi

GitHub, Jira

Oracle Database

EDUCATION

KL University

B.Tech – Electronics and Communication Engineering

CGPA: 8.67

2019 – 2023

PROJECTS

Camera Module Interfacing with Raspberry Pi

- Designed and developed a real-time surveillance and monitoring system by interfacing a camera module with Raspberry Pi. The solution was built to enhance security by capturing live visual data and enabling motion-based monitoring.
- Utilized Raspberry Pi for image acquisition, processing, and data transmission to ensure continuous system responsiveness. I have used Raspberry Pi 3 Model B which comes with 64 bit quad core processor, on board WiFi and Bluetooth and USB features and it compatible with the resolutions and the camera module can be used to set up security systems.
- The mainstream for capturing and monitoring the Visual data is Raspberry Pi Camera Module v2 which Sony IMX219 sensor. It captures static images at 3280 x 2464 pixels and supports 1080p video at 30fps. It introduces rapid autofocus (PDAF) and HDR (High Dynamic Range) support for vibrant images even in high-contrast scenes.

Gesture Controller Gaming Setup

- Developed a gesture-based gaming control system that enables interaction with digital games using hand movements instead of traditional input devices such as keyboards or controllers.
- In my project i built a gesture controller to interact with games using hand movements and gestures instead of traditional input devices like controllers or keyboards.
- The system was built using an Arduino Leonardo board integrated with flex sensors and an accelerometer to capture finger bending and hand motion data. This project involved hardware interfacing, real-time motion sensing, and gesture-to-command mapping using embedded control logic.

CERTIFICATIONS

- Introduction to Power Electronics - Coursera (2020)
- Python Programming for the Raspberry Pi - Coursera (2020)
- Multi Cloud Networking Associate – Aviatrix (2021)
- Data Science and Advanced IoT Analytics with ML – Entuple (2021, 2022)
- System Administrator – ServiceNow (2022)
- Java, SQL and Linux Technologies – FrontLineMedia (2024)